Nigel Doering

PhD Student in Data Science • UC San Diego
Halıcıoğlu Data Science Institute (HDSI) • nigel.doering@gmail.com • (231) 920-5735

Education

University of California, San Diego
Halıcıoğlu School of Data Science and Computing — Ph.D. in Data Science 2024 – present

- Advisor: Tauhidur Rahman (MOSAIC Lab) Co-Advisor: David Danks

University of California, San Diego
Halıcıoğlu Data Science Institute — M.S. in Data Science 2021 – 2023

- Advisor: David Danks

University of California, San Diego
Halıcıoğlu Data Science Institute — B.S. in Data Science 2017 – 2021

Publications

Doering, N. F.; Norman, J.; Danks, D.; and Rahman, T. 2026. *Arguing Ethics with a Machine: An AI Chatbot's Ethical Position Is a Property of the System, Not the Model*. Under review, ACM CHI Conference on Human Factors in Computing Systems (CHI 2027).

Doering, N. F.; Malladi, R.; Sangwan, A.; Tonmoy, M. T. H.; Doan, J.; Danks, D.; and Rahman, T. 2026. *Structured Machine Theory of Mind from Agent Trajectories*. Transactions on Machine Learning Research (TMLR). [OpenReview] [Code].

Doering, N. F.; Malladi, R.; Sangwan, A.; Danks, D.; and Rahman, T. 2026. *HiVAE: Hierarchical Latent Variables for Scalable Theory of Mind*. In Proceedings of the Theory of Mind for AI Workshop at the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26). Singapore. [arXiv].

Oshim, M. F. T.; Doering, N.; Islam, B.; Weng, T.-W.; and Rahman, T. 2025. *Anti-Sensing: Defense against Unauthorized Radar-based Human Vital Sign Sensing with Physically Realizable Wearable Oscillators*. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2025), 1990–1996. doi:10.1109/ICRA55743.2025.11127941. [arXiv].

Doering, N. F.; and Norman, J. D. 2025. *A FASPE Perspective on LLM-Based Chatbots*. FASPE Journal 2025 (Fellowships at Auschwitz for the Study of Professional Ethics), Design & Technology, 64–75. [PDF].

Doering, N. 2023. *Master's Thesis: Towards Anomalous Group Discovery in Maritime AIS Data*. UC San Diego eScholarship. [eScholarship].

Doering, N.; Gorlla, C.; Tuttle, T.; and Vijay, A. 2023. *Empirical Analysis of Efficient Fine-Tuning Methods for Large Pre-Trained Language Models*. arXiv preprint arXiv:2401.04051. [arXiv].

Works in Progress

Tonmoy, M. T. H.; Doering, N. F.; Parker, I.; Bharadwaj, R.; and Rahman, T. *Neuro-Mechanistic Latent Dynamics: Cognitive Load as a Generalizable Neural Stability Margin*. Manuscript in revision for resubmission.

Research & Professional Experience

FICO San Diego
Analytic Scientist Intern Summer 2026

- Conducted applied research on cybersecurity problems as part of the analytics research team.
- Developed curriculum for a course taught at Historically Black Colleges and Universities (HBCUs) in Fall 2026.
- Mentored HBCU student interns from the prior year's cohort.

Naval Research Laboratory Remote
Research Fellow June 2025 – September 2025

- Selected for the competitive NRL Fellowship to advance machine learning for operational sensing.
- Designed a framework for latent goal inference from trajectory and sensor data using theory-of-mind–inspired models with graph neural networks and world models.

Naval Information Warfare Center Pacific San Diego, CA
Data Scientist August 2021 – May 2025

- Led research streams in operational ML deployment and applied anomaly detection, bridging software engineering with novel machine learning methods for defense applications.
- Principal Investigator on a $90K ONR-funded project developing graph-based models for maritime anomaly detection.
- Designed and deployed containerized ML services for real-time operational analytics, field-tested during RIMPAC 2022.

Grants

Naval Innovation Science and Engineering (NISE) Grant 2022 – 2023
Principal Investigator — $90,000 award for research on anomaly detection in maritime AIS data.

Fellowships & Awards

- National Defense Science and Engineering Graduate (NDSEG) Fellowship
- Naval Research Laboratory Fellowship 2025
- Fellowship at Auschwitz for the Study of Professional Ethics (FASPE) — Design & Technology Cohort 2025
- Department of the Navy Civilian Service Achievement Medal 2023

Teaching

University of California, San Diego
Teaching Assistant, DSC 260 (Data Science, Ethics, and Society) Spring 2026

- Foundational concepts (power, justice, bias, privacy, explainability), societal practices (delegation, organizational incentives, accountability), and governance mechanisms (law, regulation, norms).

University of California, San Diego
Teaching Assistant, DSC 80 (Practice & Application of Data Science) Fall 2025